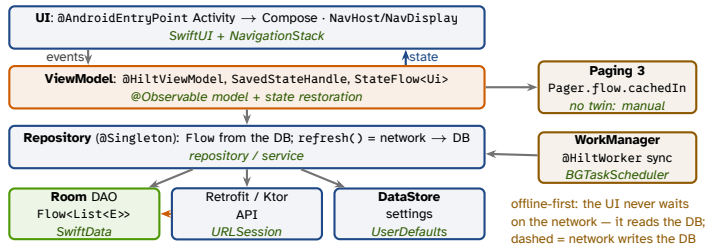
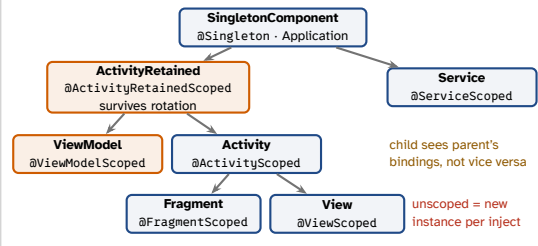


In one line: Google's recommended app: **UI** (Compose) observes **StateFlow** from a **ViewModel**, which calls a **repository** whose **single source of truth is the local DB** (Room) — the network only *writes* into it. **Hilt** builds the graph at **compile time**; **WorkManager** runs deferrable work that must survive process death and reboot. Every piece has an **iOS twin** (green).

Layers (arrow = depends on) · **iOS twin**



Hilt components (parent → child) · scope



Hilt (Dagger underneath)

- `@HiltAndroidApp` Application → `@AndroidEntryPoint` Activity/Service → `@HiltViewModel` + `@Inject` constructor. A missing binding **fails the build** (Koin fails at runtime). Use **KSP**, not **kapt**.
- `@Module @InstallIn(X::class)`: `@Binds` = interface → impl, no code; `@Provides` = factory for types you don't own (Retrofit, Room). Two of one type: `@Qualifier`. Runtime args: `@AssistedInject`; tests: `@TestInstallIn` swaps a module.

ViewModel scope + SavedStateHandle

Each **nav back-stack entry** is a `ViewModelStore`: one VM per destination, cleared when popped; share one across a flow by scoping it to the graph's entry. `SavedStateHandle` = Bundle-backed map that survives **process death** (`getStateFlow(k, d)`, `toRoute<T>()`) — ids, not data.

Room

- **SQL verified at compile time**. A Flow query re-emits on any write to its **table** — `distinctUntilChanged()`. `@Transaction` for multi-query reads (`@Relation`); `withTransaction{}` for writes.
- Migrations: `Migration(1, 2) { execSQL(...) }` or `@AutoMigration` (needs the exported schema JSON in git) + `MigrationTestHelper`; `fallbackToDestructiveMigration()` = data loss. KMP since **2.7**.

Example — the stack in 13 lines

```
@Dao interface NoteDao {
    @Query("SELECT * FROM Note") fun all(): Flow<List<Note>>
    @Upsert suspend fun upsert(n: List<Note>)
}
@Singleton class NoteRepo @Inject constructor(
    private val dao: NoteDao, private val api: NoteApi) {
    val notes = dao.all() // UI reads the DB
    suspend fun refresh() = dao.upsert(api.fetch()) // network -> DB
}
@HiltViewModel class NotesVM @Inject constructor(
    saved: SavedStateHandle, repo: NoteRepo) : ViewModel() {
    val folder = saved.toRoute<Notes>().folder // typed nav arg
    val ui = repo.notes.stateIn(viewModelScope,
        SharingStarted.WhileSubscribed(5_000), emptyList())
}
```

WorkManager — guaranteed, not punctual

- Persisted in its own SQLite DB: survives death and **reboot**, honours Doze. `CoroutineWorker` → `Result.success/failure/retry` (backoff exponential, 30 s).
- Constraints (network, charging, battery/storage not low, idle); periodic $\geq$ **15 min**. `enqueueUniqueWork(name, KEEP/REPLACE/APPEND, r)` stops duplicate syncs; `beginWith(a).then(b)` chains.
- **Expedited** (`setExpedited(OutOfQuotaPolicy...)`): quota by standby bucket; below Android 12 runs as a foreground service (`getForegroundInfo`). >10 min: `setForeground()`. Not for exact alarms or “now, visibly”.

DataStore (replaces SharedPreferences)

Preferences (keys) or **typed** (Proto / JSON + `Serializer<T>`); reads `Flow<T>`, writes `edit{}/updateData{}` — suspend, **transactional**. `SharedPreferences`: sync reads on main, `apply()` blocks in fsync. **One instance per file per process** or `IllegalStateException`.

Navigation & Paging

- **Nav Compose 2.8+**: `@Serializable` routes, `composable<Profile>`, `navigate(Profile(id))`, `toRoute()` — no string routes. **Navigation 3** (1.0 Nov 2025): **you own the back stack** (`SnapshotStateList<NavKey>`), `NavDisplay` + `entryProvider` — like `NavigationStack(path:)`.
- **Paging 3**: `PagingSource.load()` → `LoadResult.Page(data, prevKey, nextKey)`; `Pager.flow.cachedIn(scope)`; `RemoteMediator` fills Room from the network.

Traps

- ! `@Singleton` everywhere — state leaks across logins.
- ! “WorkManager runs it now” — *eventually*, once constraints hold.
- ! A `DataStore` created in a composable — two instances, crash.

Remember: UI ← VM ← repo ← DB; the network writes the DB · Hilt at compile time · WorkManager survives reboot.